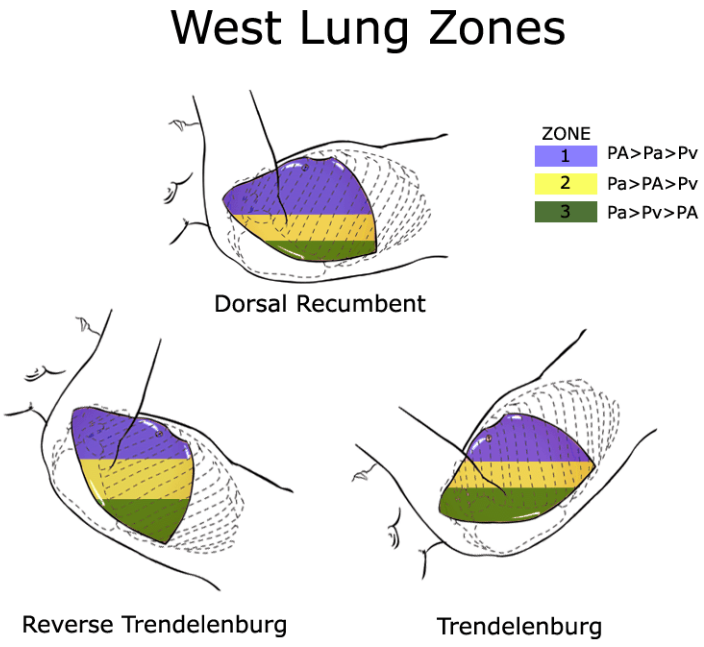
West et al, defined three lung zones based on the relationship between pressures in the alveoli, arterioles, and venules.

In Zone I, the pressure in the alveoli exceeds that of the arterioles and the venules. Zone 1 is rarely present normally, but can occur in pulmonary hypotension, excessive PEEP, or large tidal volumes during positive-pressure ventilation.

In Zone 2, the pressure in the arterioles exceeds that of the alveoli and the venules. This occurs in the nondependent portions of the lung and perfusion is the result of the oscillating balance between arterial and alveolar pressures.

In Zone 3, the pressure caused by hydrostatic forces (often due to gravity) produce venous congestion and perfusion is dictated by the difference in arterial pressure and venous pressure.



The illustration to the right demonstrates the changes that occur with patient positioning. Because of gravity, Zone III remains the most dependent area, such that in the head up position, Zone III shifts more toward the lung bases and in the head down position, Zone III shifts into the apex which is then the most dependent position.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 274.

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